

Trigonometry — Complete Notes

Grade 12

Mathematics

CAPS

What you'll learn

- The special-angle values you must know by heart.
- The compound angle formulae for sine, cosine and tangent.
- The double angle formulae and how they are derived.
- How to write general solutions for trigonometric equations.
- How to prove trigonometric identities step by step.
- How to simplify expressions using reduction formulae.

The big picture

Grade 12 trigonometry extends what you learned in earlier grades. In Grades 10 and 11 you solved simple equations like $\sin\theta = 1/2$. In Grade 12 the angles become more complicated: sums and differences of angles, double angles, and equations that need a **general solution** because they repeat forever.

The whole chapter rests on a handful of formulae. Learn them, then practise proving identities — proving an identity is a favourite CAPS question because it tests both memory and logic.

Special angles

Know these values exactly — no calculator needed:

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undefined

For signs by quadrant, remember **CAST**: All ratios positive in the first quadrant, only Sine in the second, only Tangent in the third, only Cosine in the fourth.

Fundamental identities

- $\sin^2\theta + \cos^2\theta = 1$ (the Pythagorean identity)
- $\tan\theta = \sin\theta / \cos\theta$

From the first identity:

- $\sin^2\theta = 1 - \cos^2\theta$
- $\cos^2\theta = 1 - \sin^2\theta$

Compound angle formulae

These express the sine, cosine and tangent of a sum or difference of two angles:

- $\sin(A + B) = \sin A \cdot \cos B + \cos A \cdot \sin B$

- $\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$
- $\cos(A + B) = \cos A \cdot \cos B - \sin A \cdot \sin B$
- $\cos(A - B) = \cos A \cdot \cos B + \sin A \cdot \sin B$
- $\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \cdot \tan B)$
- $\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \cdot \tan B)$

Notice the **sign flip**: in the sine and tangent formulae the signs on the right match the sign inside the bracket, but in the cosine formula the sign on the right is **opposite** to the sign in the bracket.

Double angle formulae

Set $B = A$ in the compound formulae:

- $\sin 2A = 2 \cdot \sin A \cdot \cos A$
- $\cos 2A = \cos^2 A - \sin^2 A$
- $\cos 2A = 1 - 2\sin^2 A$
- $\cos 2A = 2\cos^2 A - 1$
- $\tan 2A = 2\tan A / (1 - \tan^2 A)$

The three forms of $\cos 2A$ are all the same thing — choose the one that helps you cancel or substitute.

General solutions

Because trigonometric functions repeat, equations usually have infinitely many solutions. We give a **general solution** using an integer n :

Equation	General solution
$\sin \theta = k$	$\theta = \arcsin(k) + 360^\circ n$ or $\theta = 180^\circ - \arcsin(k) + 360^\circ n$
$\cos \theta = k$	$\theta = \pm \arccos(k) + 360^\circ n$
$\tan \theta = k$	$\theta = \arctan(k) + 180^\circ n$

If you must solve within a restricted interval (for example $0^\circ \leq \theta \leq 360^\circ$), generate values by choosing $n = 0, \pm 1, \pm 2, \dots$ and keep only those inside the interval.

Sine and cosine repeat every 360° , while tangent repeats every 180° — that is exactly why the "+ $360^\circ n$ " and "+ $180^\circ n$ " appear.

Worked example 1 — exact value of a compound angle

Find $\sin 75^\circ$ without a calculator.

Step 1 — split 75° into known angles. $75^\circ = 45^\circ + 30^\circ$.

Step 2 — apply the sine addition formula. $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cdot \cos 30^\circ + \cos 45^\circ \cdot \sin 30^\circ$

Step 3 — substitute the special values. $= (\sqrt{2}/2)(\sqrt{3}/2) + (\sqrt{2}/2)(1/2) = (\sqrt{6})/4 + (\sqrt{2})/4 = (\sqrt{6} + \sqrt{2}) / 4$

So $\sin 75^\circ = (\sqrt{6} + \sqrt{2})/4 \approx 0,966$.

Worked example 2 — general solution

Solve $\cos 2x = 1/2$ for the general solution.

Step 1 — take the inverse cosine. $\arccos(1/2) = 60^\circ$.

Step 2 — apply the cosine general solution. $2x = \pm 60^\circ + 360^\circ n$

Step 3 — divide every part by 2. $x = \pm 30^\circ + 180^\circ n$

So $x = 30^\circ + 180^\circ n$ or $x = -30^\circ + 180^\circ n$, which can also be written as $x = 150^\circ + 180^\circ n$.

Check: for $n = 0$ we get 30° and 150° ; doubling each gives 60° and 300° , and $\cos 60^\circ = \cos 300^\circ = 1/2$. Correct.

Worked example 3 — prove an identity

Prove that $\sin 2A / (1 + \cos 2A) = \tan A$.

Step 1 — work on the left-hand side only. LHS = $\sin 2A / (1 + \cos 2A)$

Step 2 — replace with double angle formulae. $\sin 2A = 2 \cdot \sin A \cdot \cos A$ and $\cos 2A = 2\cos^2 A - 1$, so: $1 + \cos 2A = 1 + (2\cos^2 A - 1) = 2\cos^2 A$

Step 3 — substitute back and simplify. LHS = $(2 \cdot \sin A \cdot \cos A) / (2\cos^2 A) = \sin A / \cos A$

Step 4 — recognise the identity. $\sin A / \cos A = \tan A = \text{RHS} \therefore$ the identity is proven.

Reduction formulae

Angles bigger than 90° , and negative angles, can all be reduced to an acute angle between 0° and 90° using these rules:

Expression	Simplified
$\sin(180^\circ - \theta)$	$\sin \theta$
$\cos(180^\circ - \theta)$	$-\cos \theta$
$\tan(180^\circ - \theta)$	$-\tan \theta$
$\sin(180^\circ + \theta)$	$-\sin \theta$
$\cos(180^\circ + \theta)$	$-\cos \theta$
$\tan(180^\circ + \theta)$	$\tan \theta$
$\sin(360^\circ - \theta)$	$-\sin \theta$
$\cos(360^\circ - \theta)$	$\cos \theta$
$\tan(360^\circ - \theta)$	$-\tan \theta$
$\sin(90^\circ - \theta)$	$\cos \theta$
$\cos(90^\circ - \theta)$	$\sin \theta$
$\sin(-\theta)$	$-\sin \theta$
$\cos(-\theta)$	$\cos \theta$
$\tan(-\theta)$	$-\tan \theta$

Two patterns make the table easy to remember. For 180° and 360° the function name never changes; only the sign may change, and that sign follows CAST. For 90° and 270° the function name swaps between sine and cosine, because these are complementary angles.

Example: simplify $\sin(180^\circ + x) \cdot \tan(360^\circ - x)$. $\sin(180^\circ + x) = -\sin x$ and $\tan(360^\circ - x) = -\tan x$, so the product = $(-\sin x)(-\tan x) = \sin x \cdot \tan x = \sin^2 x / \cos x$.

Choosing a method for equations

When an equation contains a double angle together with a single angle (for example $\sin 2x = \sin x$), factorise rather than reaching straight for a general-solution formula. Move everything to one side, take out the common ratio as a factor, and solve each factor separately. This avoids losing solutions that would otherwise be divided away.

Common mistakes

- **Sign errors in the cosine formula.** $\cos(A + B)$ has a **minus** in its expansion — the sign flips.
- **Forgetting the "+ $360^\circ n$ ".** A general solution without the period term is incomplete and loses marks.
- **Working on both sides of an identity.** Prove one side until it matches the other; do not move terms across the equals sign as if solving an equation.
- **Using the wrong form of $\cos 2A$.** All three forms are valid, but only one may cancel nicely in a given proof.
- **Mixing up degrees and radians.** Unless told otherwise, CAPS uses degrees.
- **Dropping solutions when dividing an angle.** If $2x = \pm 60^\circ + 360^\circ n$, remember to divide the whole right side by 2 to get $\pm 30^\circ + 180^\circ n$.

Quick summary / cheat sheet

Formula type	Result
Pythagorean	$\sin^2 \theta + \cos^2 \theta = 1$
$\sin(A \pm B)$	$\sin A \cdot \cos B \pm \cos A \cdot \sin B$
$\cos(A \pm B)$	$\cos A \cdot \cos B \mp \sin A \cdot \sin B$
$\tan(A \pm B)$	$(\tan A \pm \tan B) / (1 \mp \tan A \cdot \tan B)$
$\sin 2A$	$2 \cdot \sin A \cdot \cos A$
$\cos 2A$	$\cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\tan 2A$	$2\tan A / (1 - \tan^2 A)$

Practice questions

1. Find the exact value of $\cos 15^\circ$.
2. Give the general solution of $\tan 2x = 1$.
3. Prove that $1 - \cos 2A = 2\sin^2 A$.
4. Solve $\sin 2x = \sin x$ for x in $[0^\circ; 360^\circ]$.

5. If $\sin A = 3/5$ and A is acute, find $\sin 2A$ and $\cos 2A$.

6. Simplify $\sin(180^\circ - x) \cdot \cos(90^\circ + x)$.

Answers

1. $\cos 15^\circ = \cos(45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = (\sqrt{6} + \sqrt{2})/4 \approx 0,966$.

2. $2x = 45^\circ + 180^\circ n$, so $x = 22,5^\circ + 90^\circ n$.

3. $\text{LHS} = 1 - (1 - 2\sin^2 A) = 1 - 1 + 2\sin^2 A = 2\sin^2 A = \text{RHS} \therefore$ proven.

4. $2\sin x \cdot \cos x = \sin x \rightarrow \sin x(2\cos x - 1) = 0$. So $\sin x = 0 \rightarrow x = 0^\circ, 180^\circ, 360^\circ$; or $\cos x = 1/2 \rightarrow x = 60^\circ, 300^\circ$.

Full solution set: $x = 0^\circ, 60^\circ, 180^\circ, 300^\circ, 360^\circ$.

5. $\cos A = 4/5$ (a 3-4-5 triangle). $\sin 2A = 2(3/5)(4/5) = 24/25$. $\cos 2A = 1 - 2\sin^2 A = 1 - 2(9/25) = 1 - 18/25 = 7/25$.

6. $\sin(180^\circ - x) = \sin x$ and $\cos(90^\circ + x) = -\sin x$, so the product = $-\sin^2 x$.

Official EduRank Team notes. Always cross-check with your textbook and teacher.